

Lab 2 - Collections in Python

Part A - Lists

1. Create a list of at least eight programming languages. Access the first, last, third and second-to-last values.
2. Print three different slices of the list, then print the list in reverse order using slicing.
3. Use append, insert, remove and pop. After each operation, print the list so the change is visible.
4. Create a numeric list. Calculate its length, minimum, maximum and sum using built-in functions.
5. Sort one list ascending and another descending. Explain in a comment the difference between changing a list with `.sort()` and creating a sorted result with `sorted()`.
6. Demonstrate the reference/copy issue using `list_b = list_a`. Then fix it with `.copy()`

Part B - Tuples and unpacking

1. Create a tuple representing RGB values. Unpack it into three variables and print them.
2. Create a tuple containing a person's name, age and city. Unpack and use the values in a formatted sentence.
3. Attempt to reason about changing one tuple element. Explain in a comment why tuples are useful when values should not be changed.
4. Create a list containing at least four coordinate tuples such as `(10, 20)`. Access individual x and y values.

Part C - Sets

1. Create a list containing duplicate course names. Convert it to a set and compare the lengths before and after.
2. Create two sets representing skills of two developers. Find skills they share, skills only the first has, and all skills represented by either person.
3. Create a set and practice add, remove/discard and membership testing.

4. Explain in comments why a set is a better choice than a list for one real-world uniqueness problem.

Part D – Dictionaries

1. Create a dictionary for a laptop with brand, model, RAM, storage and price. Read every value by key.
2. Update the price, add an operating_system key and remove one key.
3. Use .get() for both an existing and a missing key. Compare it conceptually with direct indexing.
4. Print keys, values and items separately.
5. Create a dictionary mapping five course names to number of study hours. Calculate the total hours using the dictionary values.

Part E - Nested collections

1. Create a list of at least five dictionaries representing books with title, author, pages and available.
2. Access the title of the third book and the availability of the last book.
3. Change one nested value and add a new key to one book.
4. Create a dictionary where each key is a department and each value is a list of employee names.
5. Create a structure for three courses where each course contains a name, teacher and list of topics. Print one specific topic using chained indexing.

Part F - Applied challenge: Personal media catalogue

1. Create a catalogue containing at least eight movies, games or books. Each item must be a dictionary with at least four useful fields.
2. Store all item dictionaries in one list.
3. Create a set containing all unique categories/genres represented in the catalogue.
4. Create a tuple for each item's immutable identifier plus release year, and include or associate it sensibly in your design.

5. Perform at least ten manual retrieval/update operations that demonstrate nested indexing, membership and collection methods.
6. Print a clean summary of the catalogue without using loops yet. Repetition is acceptable here because loops come next lesson.

Part G - Stretch challenges

1. Given two lists of usernames, determine duplicates and unique usernames using sets.
2. Design a nested collection for a small online course platform: courses, teacher, students and topics. Do not write classes.
3. Create a dictionary-based inventory for five products. Update stock values manually and calculate total units using values.
4. Write a short comparison in comments: list vs tuple vs set vs dictionary. Give one situation where each is the best fit.